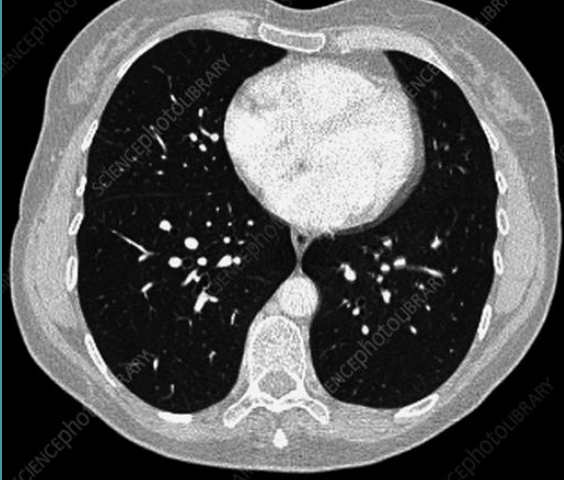




Advancing Lung Cancer Diagnostics through Multimodal Deep Learning: Optimal Fusion of 3D Radiological Imaging and Clinical Data with LLM-Driven Interpretability



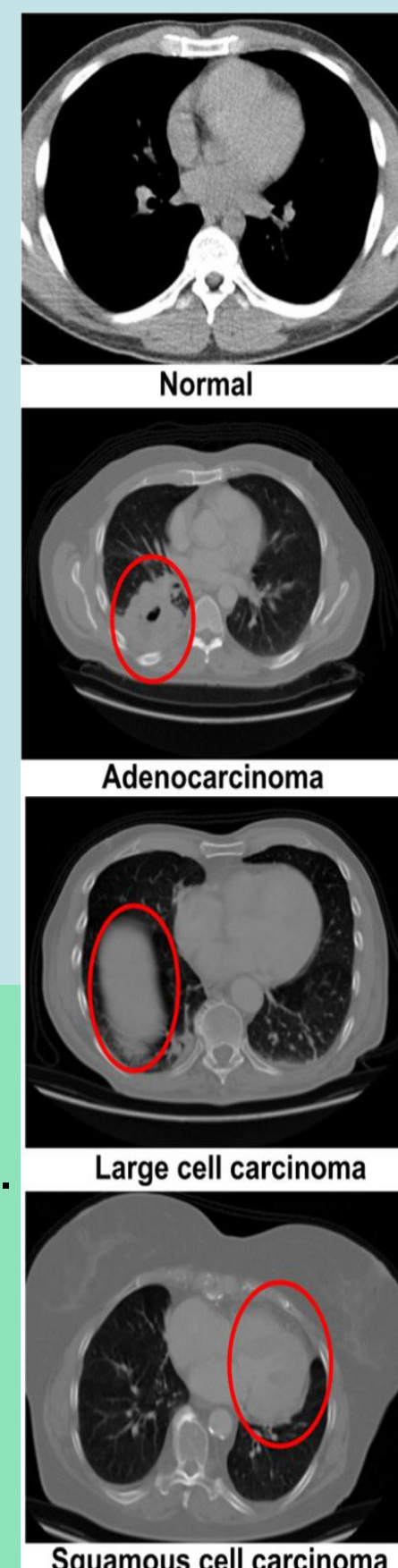
INTRODUCTION

Lung Cancer is associated with the abnormal growth of cells.

- Formation of nodules/tumors reduces normal lung function.
- Second most common cancer globally (2.2 million cases/year).
- Survival rate of only 10-20%, varies for different types.

Early Detection of Lung Cancer

- Crucial for personalized treatment.
- Improves survival rates (tumors double every 30-90 days).
- Computed Tomography (CT) based screening is useful
 - Non-invasive, high accuracy and high resolution.
 - Allows doctors to identify/assess tumor size & location.



Key Challenges in Detection

- Radiologist shortage: 1 per 1000 (underdeveloped regions).
- High misdiagnosis rates: ~30% errors in early stages.
- Delays in diagnosis: average of 90 days delay in detection.
- Expensive and time consuming: results take 2-4 weeks.

Multi-Modal Approach for Early Detection

- Multimodal approaches enable integration of diverse data types—such as imaging and demographic information to provide holistic diagnostics.
- Large Language Models (LLM) have shown promise in enhancing model interpretability by generating human-readable explanations that highlight the key reasons for why a patient is flagged as high-risk.

	Data Sources	Interpretability Radiologist's Report	Outcomes
Traditional	Medical Images		Second Opinion and Treatment
Multimodal AI	Medical Images Demographic Data Clinical Reports	Large Language Model Interpretation + Model Insights	Early and Faster Detection Personalized Treatment Global Healthcare Access

An LLM-enabled nodule/tumor diagnostics system can help pulmonologists gain confidence in promptly starting personalized treatment for their patients.

RESEARCH STUDY

GOAL: Develop and implement a novel, multimodal algorithm for robust and early detection of lung nodules/tumors which outperforms existing methods.

RESEARCH QUESTION

What is the most effective method of combining CT scans and patient data based on feature fusion to improve the accuracy and interpretability of early-stage lung cancer diagnostics?

HYPOTHESIS

If the training dataset is consistent, balanced, and sufficiently diverse, then combining features extracted by transfer-learning models, clinical reports summarized by large language models, and key demographic predictors will yield more accurate diagnostic outcomes than CT scans alone, while also providing valuable insights for precision medicine.

DESIGN SPACE

Independent Variables:

Variable	Levels/Settings
Predictor(s)	[CT only, Multimodal]
CT Feature Extraction	[3D ResNet, MedicalNet, 2D VGG16]
Classifier	[Logistic, XGBoost, Random Forest, KNN, Naïve Bayes]

Dependent Variables:

$$1. \text{Accuracy, } A = \frac{TP+TN}{TP+FP+TN+FN}$$

$$2. \text{Precision, } P = \frac{TP}{TP+FP}$$

$$3. \text{Recall, } R = \frac{TP}{TP+FN}$$

$$4. \text{F1 Score} = \frac{2 \times P \times R}{P+R}$$

Hyperparameters:

AI Algorithms	Hyperparameters	Range/Settings
ML		
K Nearest Neighbors (KNN)	n-neighbors	[1-50]
Logistic Regression (LR)	Solvers	[lbfgs, liblinear]
XGBoost (XGB)	Learning Rate	[0.01-1]
Random Forest (RFC)	Number of Estimators	[10-200]
Naïve Bayes (NB)	Smoothing (alpha)	[0.001-10]
DL & Transfer Learning		
3DResNet, MedicalNet	Number of Layers	[1-5]
	Learning Rate	[0.01-1]
VGG16	Optimizer	Adam, SGD
	Batch Size	Various
	Epochs	[1-50]

Controlled Variables: (1) CT scan resolution and preprocessing (2) Same data partitions

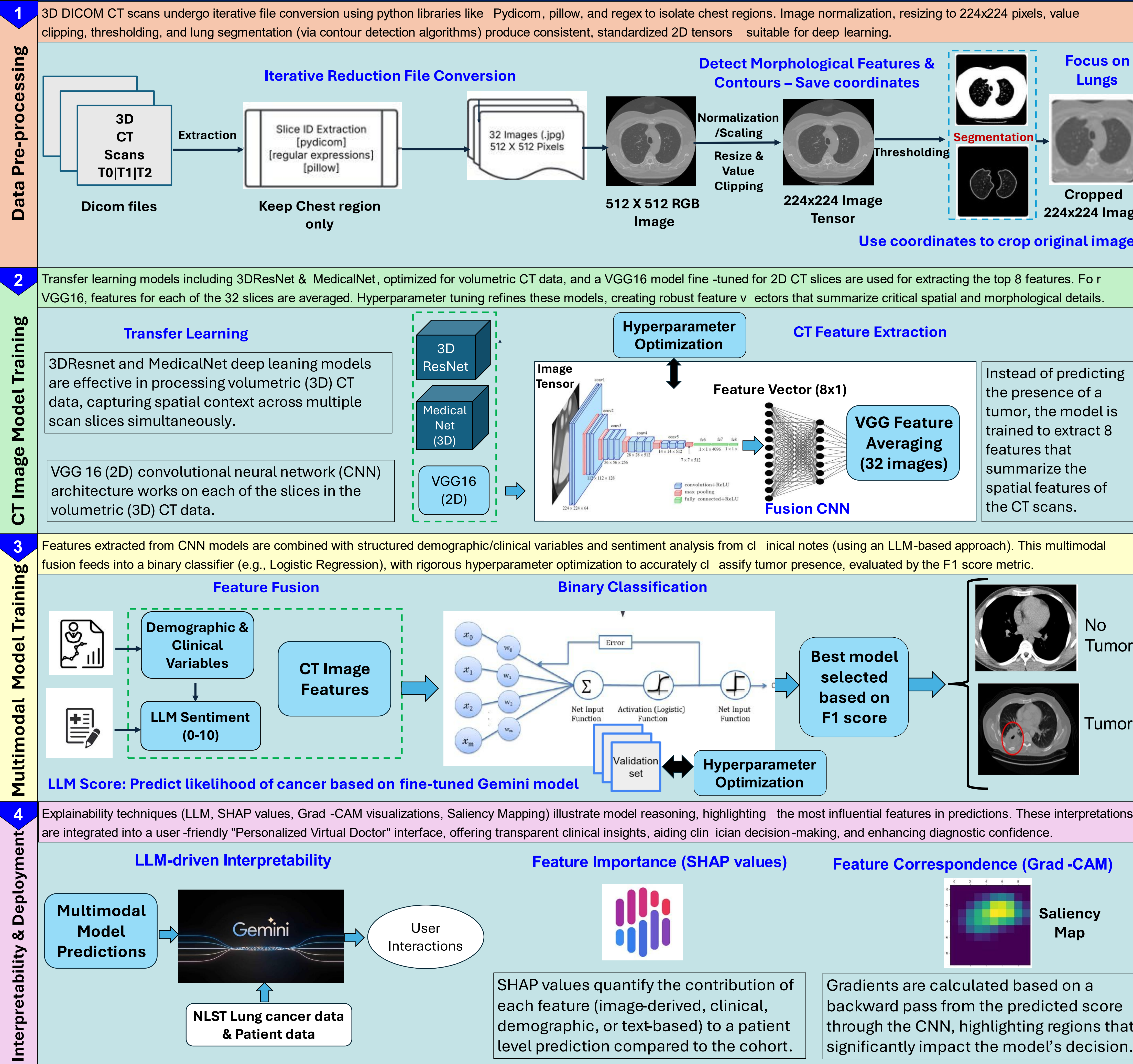
Control: Validation dataset

DATA SOURCES

- National Lung Screening Trial (NLST) dataset** from National Institute of Health (NIH) with 53,000 participants (ages 55-74), high-risk smokers:
 - Training: 310 cancer (58,042 images); 369 no cancer (50,898 images)
 - Validation: 312 cancer (58,982 images); 184 no cancer (30,519 images)
- Alongside CT scans, patient-aligned data for:
 - extensive clinical & demographic information.
 - clinical outcomes, cancer type & survival data.
 - sample variables shown.

Demographic Data	Smoking History
Age Ethnicity Gender Height Race Weight	Age Quit Smoked Cigar Smoked Pipe Pack Years Smoking Onset Age Avg Days Smoked Lived With Smoker Worked With Smoker Total Years Smoked

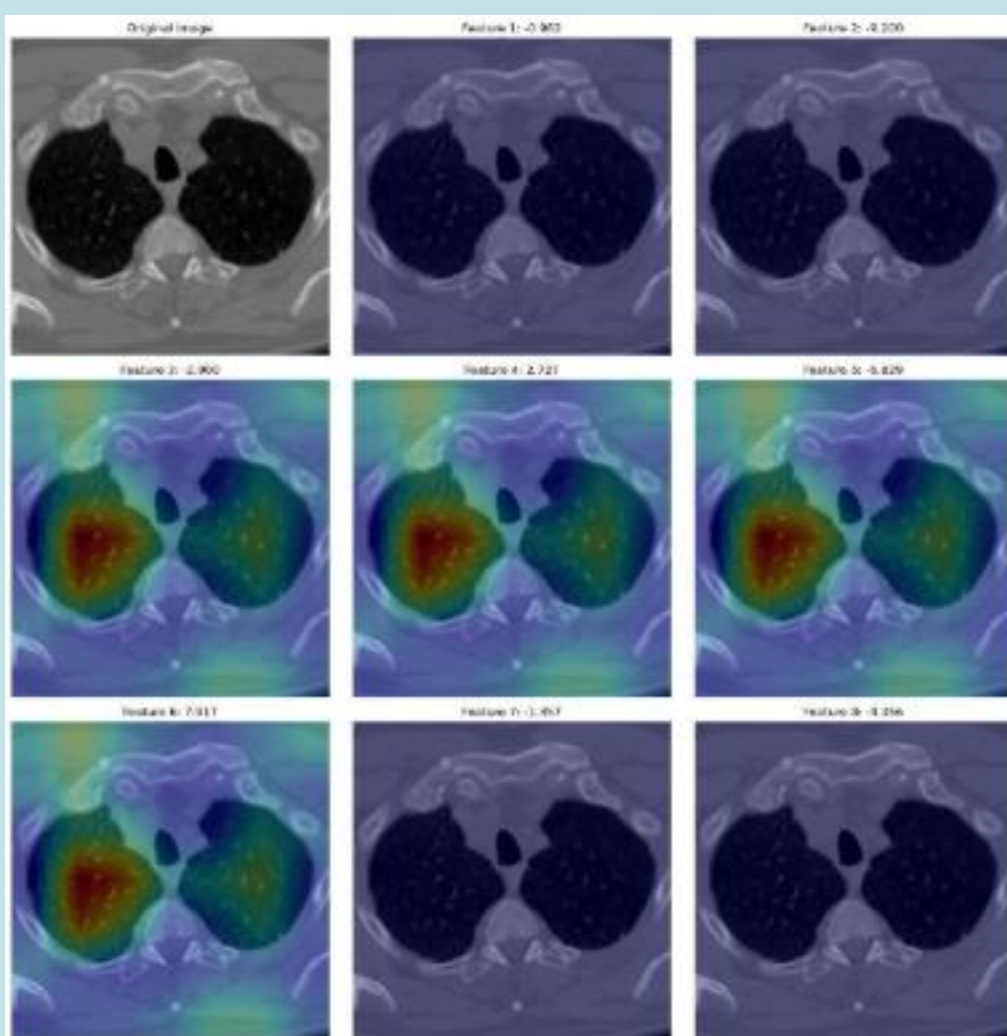
METHODOLOGY



STATISTICAL ANALYSIS and RESULTS

MODEL TRAINING AND VALIDATION

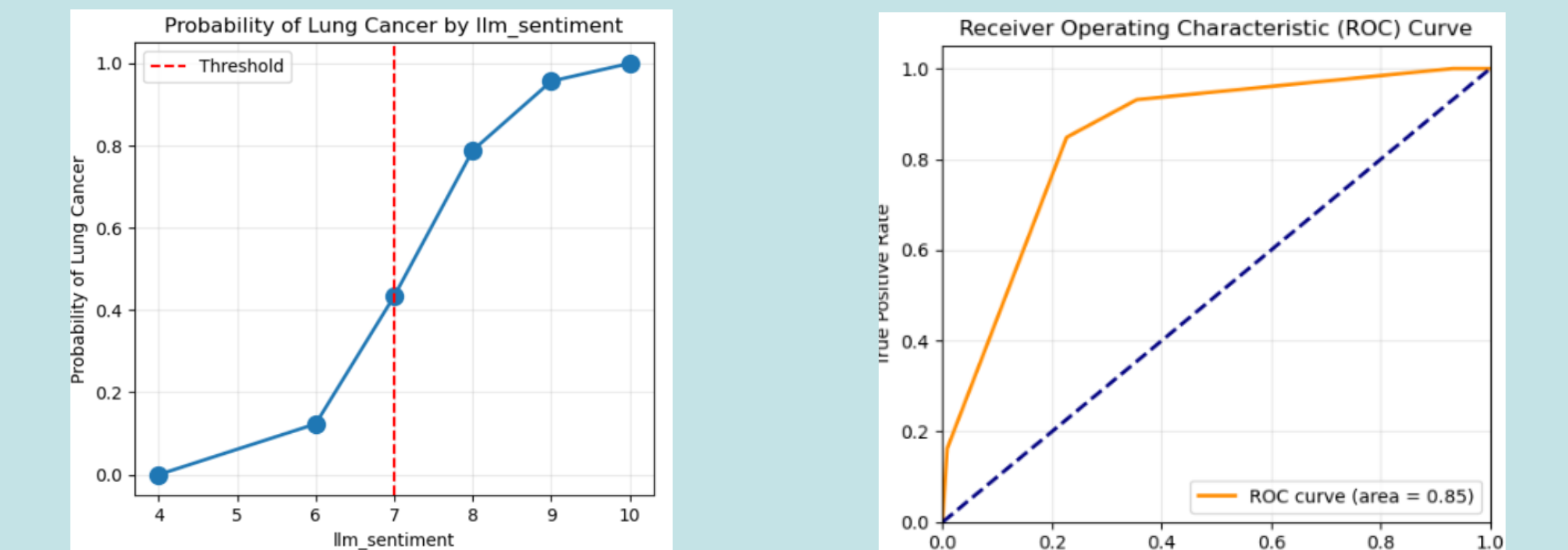
CT Feature Extraction: 2D VGG16 performs better than the 3D models.



Original image and extracted 8 features

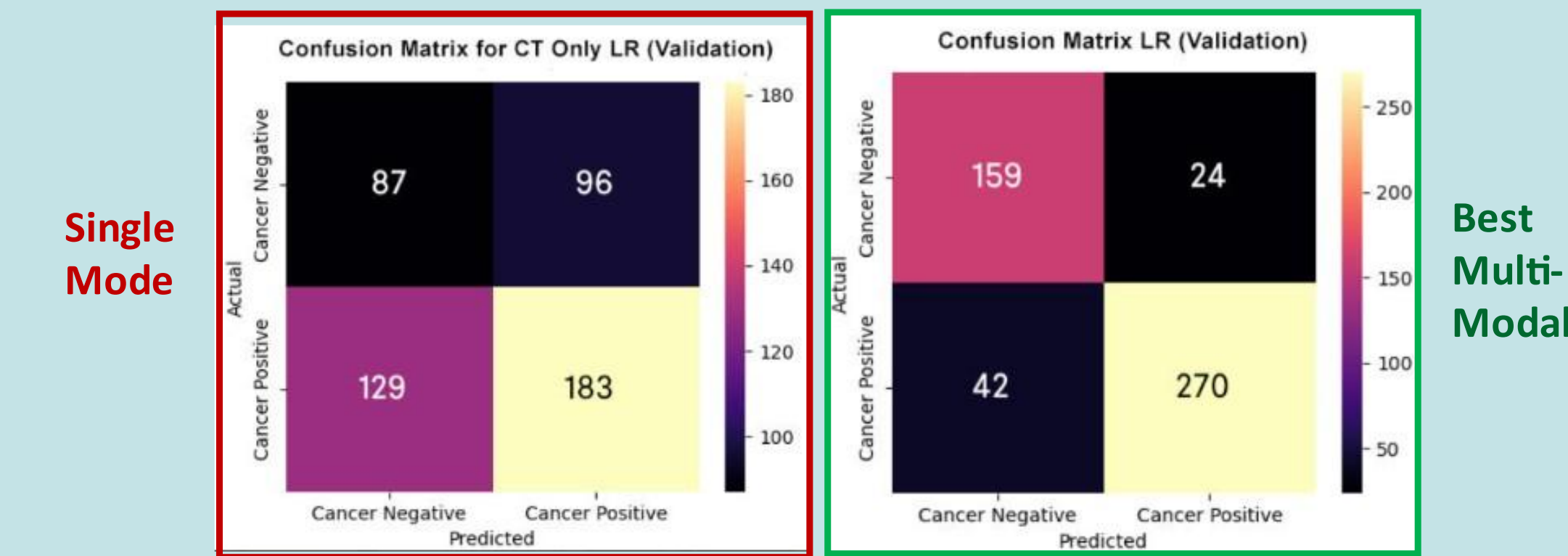
- Hotspots in Grad-CAM (Gradient-weighted Class Activation Mapping) visualizations indicate regions that strongly influence each feature's contribution to the model's prediction.
- Analysis shows that not all eight features show up as hotspots. These are dormant for a given subject.
- All features activate parts of lungs, or their edges associated with the class.

LLM Model: Lung cancer probability increases with higher LLM scores. A threshold of 7 marks a strong shift toward high-risk predictions.



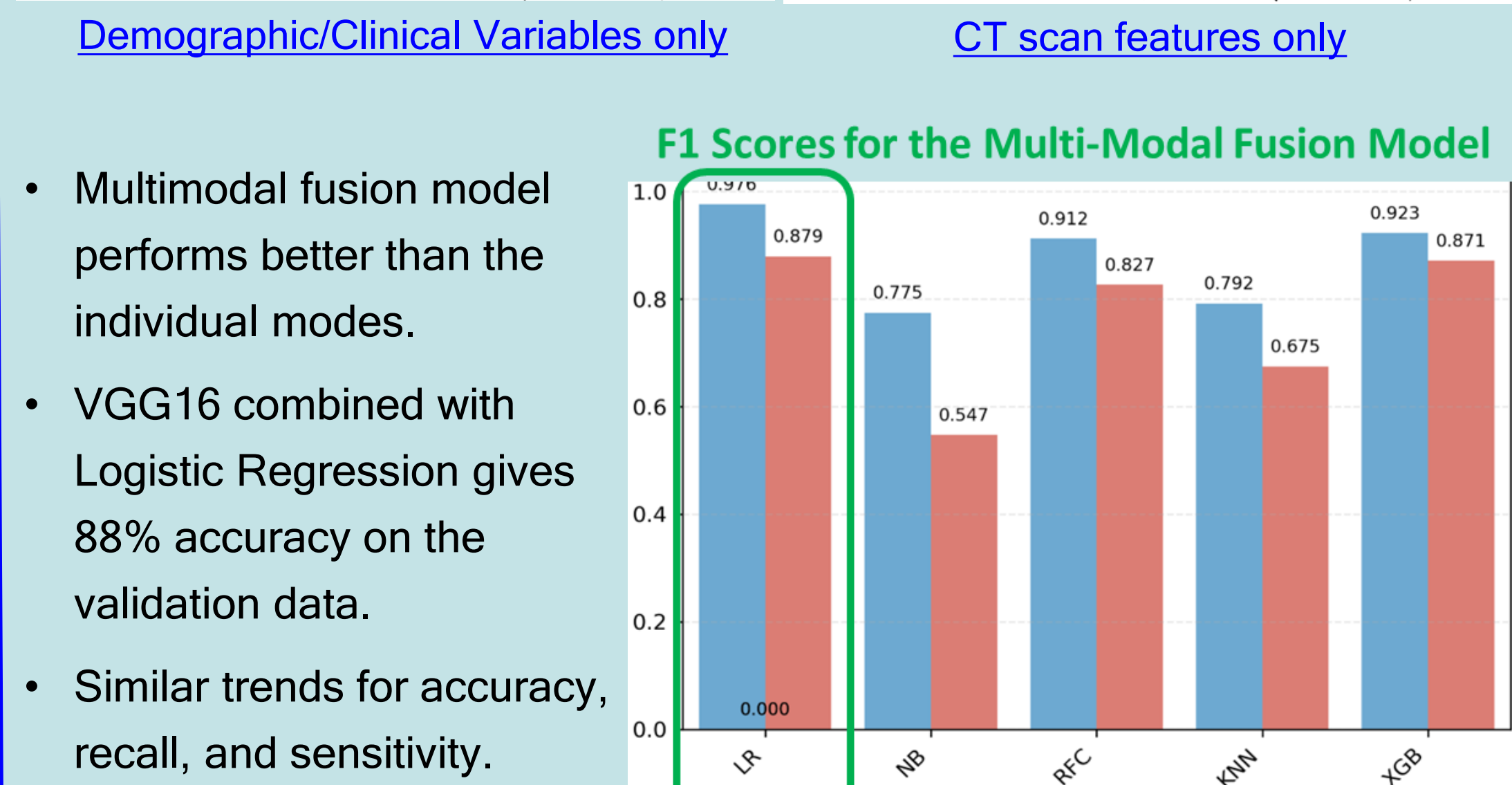
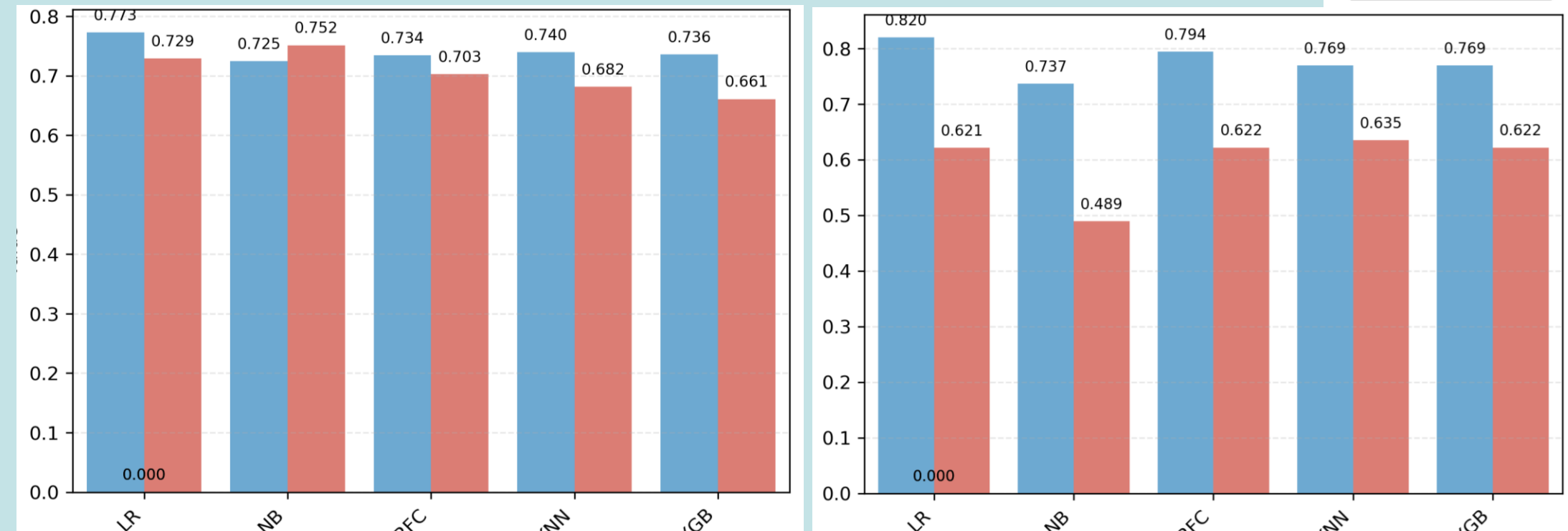
Binary Classification Models: Performance comparison for model with features only from CT against multimodal model (CT + demographic + LLM score)

CONFUSION MATRICES

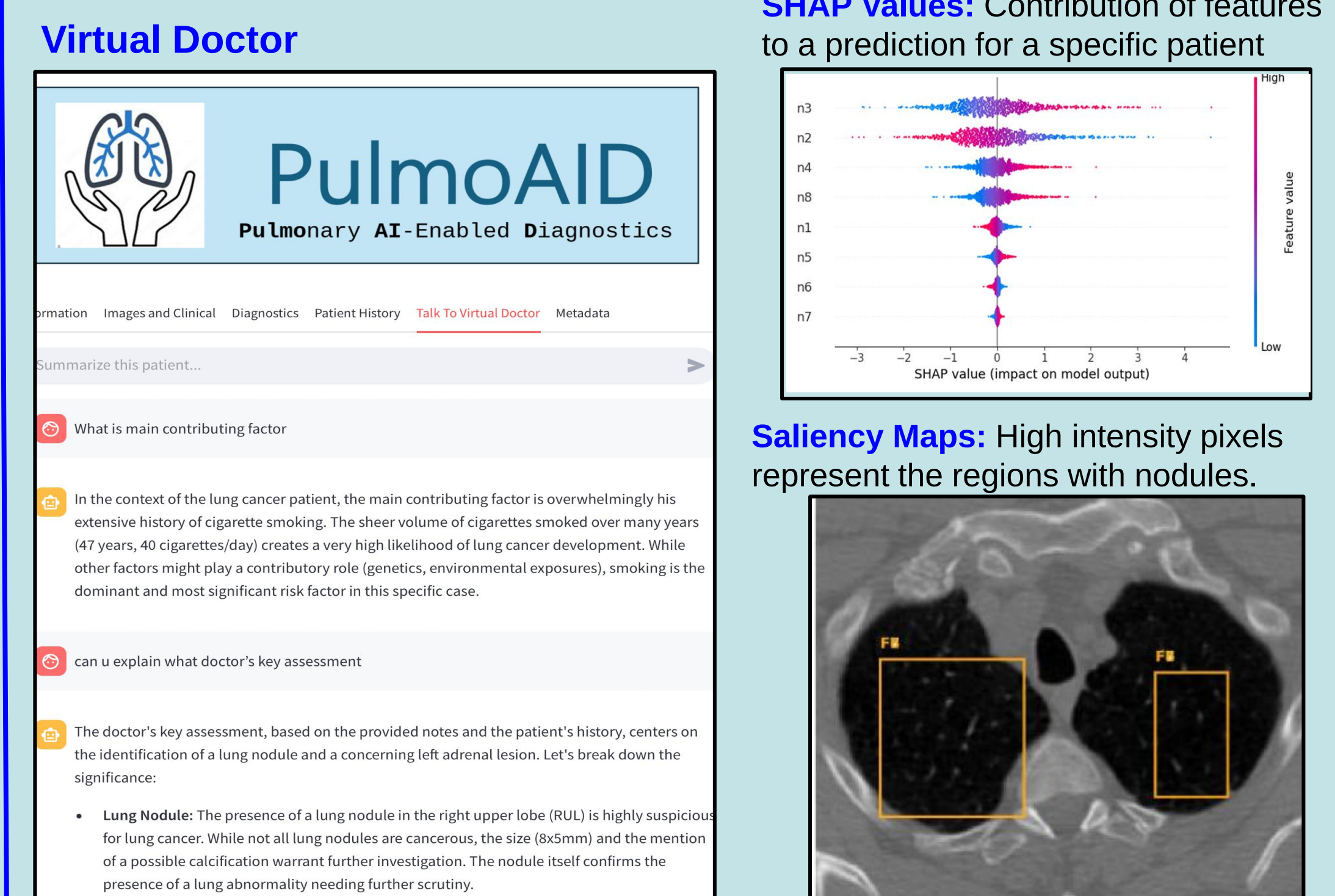


MODEL PERFORMANCE METRICS

Binary Classifier Performance: F1 scores for different models with training and validation data sets.

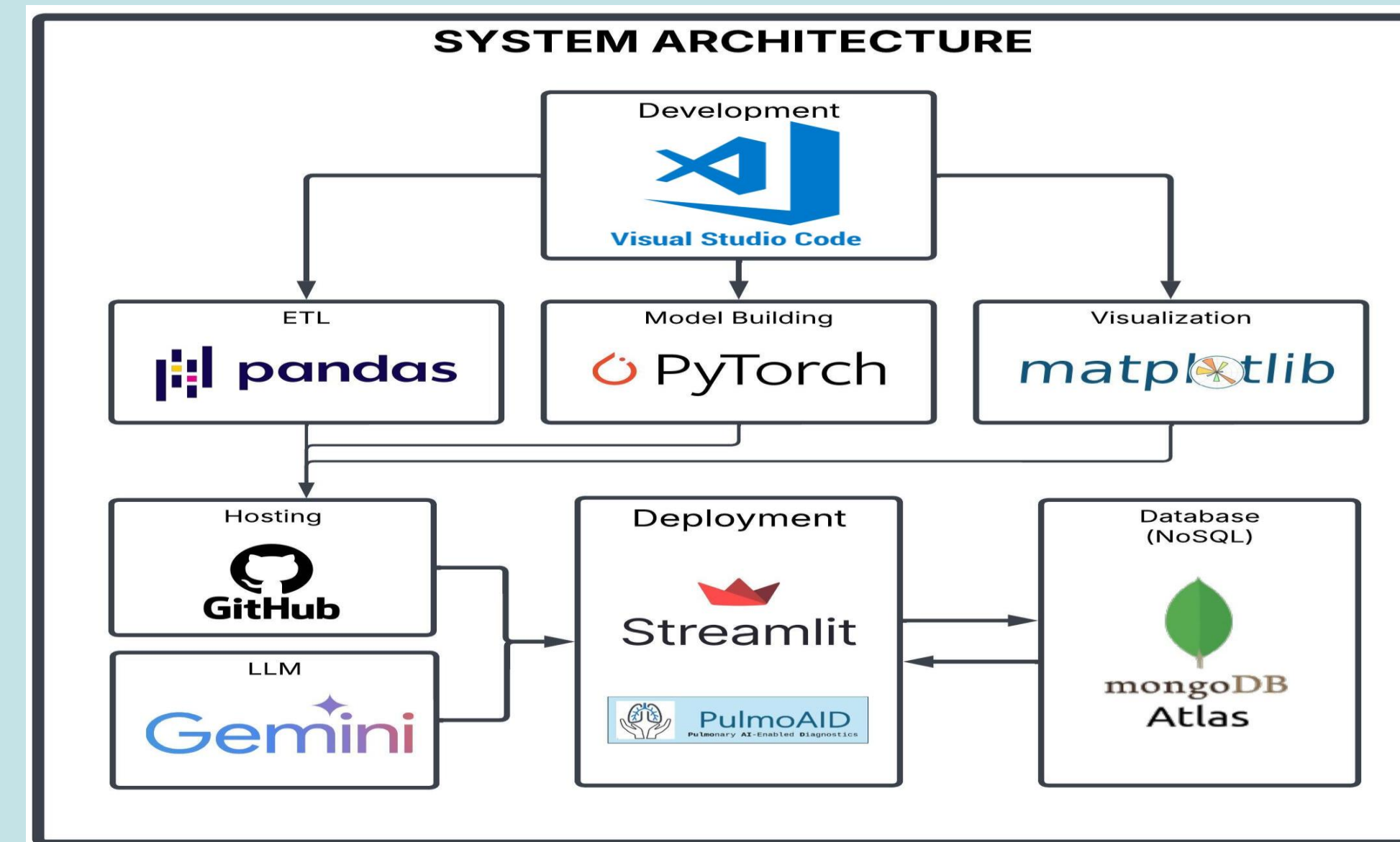


PERSONALIZED INTERPRETABILITY



WEB APPLICATION

DESIGN AND DEVELOPMENT

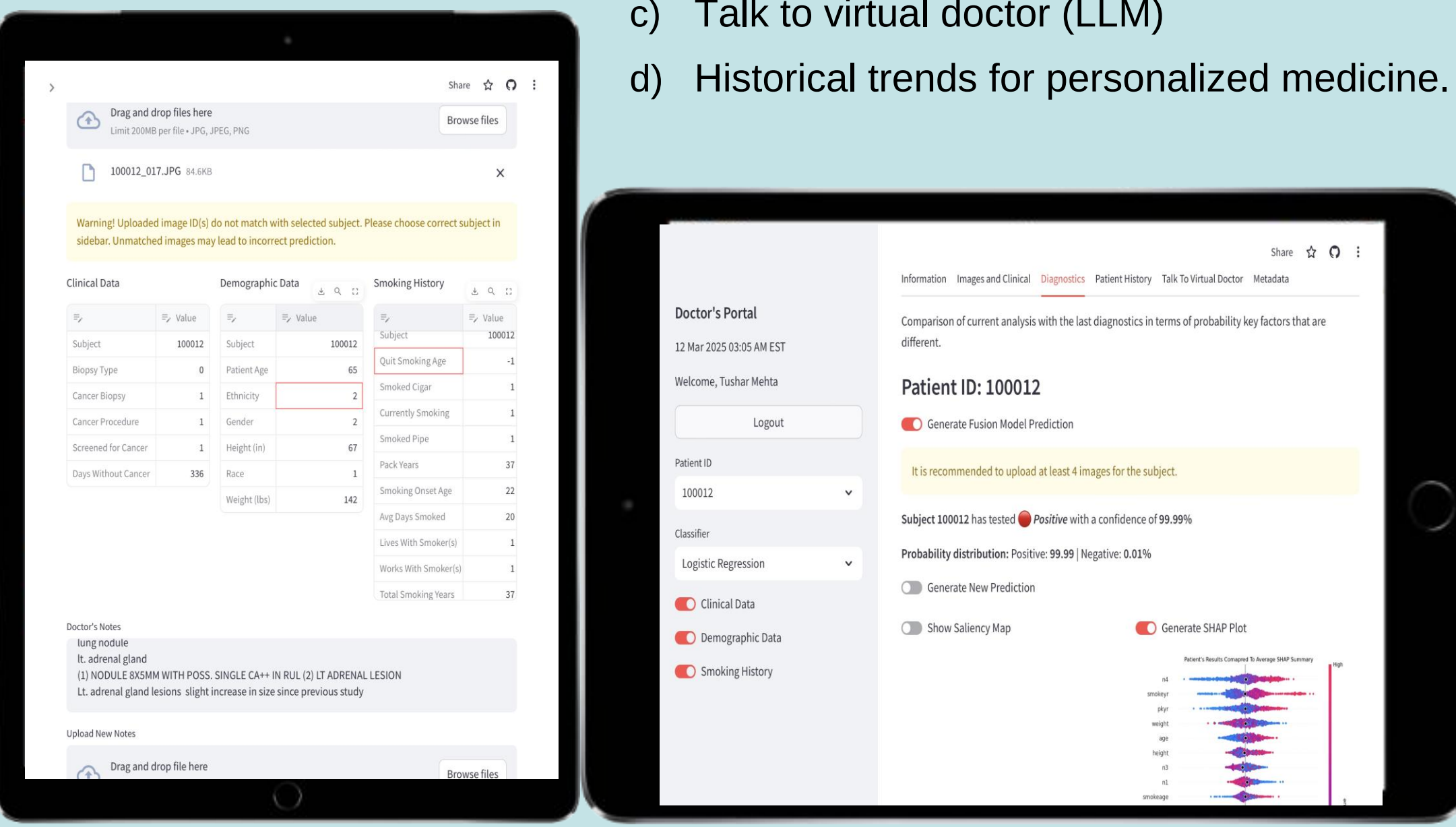


- NoSQL Database:** mongoDB was used to Stores structured and unstructured data (images, text, outcomes). Cloud API integration allows automatic connection handling—no manual setup needed.
- LLM Integration:** Leveraged Gemini's API to overcome challenges associated with computational resources, hosting, or deployment.
- Model Hosting:** Models are stored in GitHub. Binary classifiers are saved as Joblib pickle files. The larger VGG16 model is stored in chunks and loaded once during app startup for efficiency.

PulmoAID: Pulmonary AI-enabled Diagnostic System

4 step Screening Diagnosis

- Inputs:**
- Select patient ID
 - Select classifier model
 - Upload the CT scan(s)
 - Update patient data
- Outputs:**
- Probability for the presence of tumor
 - Medical insights (Explainability)
 - Talk to virtual doctor (LLM)
 - Historical trends for personalized medicine.



DISCUSSION

Conclusion: Data supported the original hypothesis that multimodal fusion of features extracted from CT scans with LLM sentiments from clinical reports and key demographic predictors has a higher F1 score (88%) compared to the model based on CT scans only (62%).

KEY RESULTS

- Proposed **novel fusion model** combining features from 2D slices of CT based on VGG16 along with demographic/clinical predictors used as the input into Logistic Regression classifier outperforms CT image models.
- 2D VGG16 transfer learning outperforms 3D CNN models (3DResNet & MedicalNet) due to the poor consistency of CT scans.
- Unlike traditional models based on CT scans, the LLM-based multi-modal method interprets nuanced medical histories, providing a comprehensive risk assessment, improving predictive accuracy and interpretability.
- SHAP values and Grad-CAM's provide insights on developing nodules, affected areas & contributing factors, not available from visual diagnostics.
- PulmoAID delivers state-of-the-art lung cancer diagnostics through a secure platform, helping democratize screening access globally.

CLINICAL APPLICATIONS

- Digital Radiologist:** Intelligent and explainable capabilities of PulmoAID provide confidence and equitable access for patients at point-of-care in under-served rural areas and under-developed countries as well as instantaneous diagnosis in emergency rooms.
- Precision Medicine:** The LLM-based virtual doctor provides human-understandable diagnostics and personalized recommendations based on patient specific data, CT features and medical research.

ONGOING AND FUTURE RESEARCH

Ideas for improvement and reducing classification error:

- Continue field trials with protected patient data & include those in training.
- Hybrid model: Optimal selection of fused features from TL models.
- Next Steps:**
 - Pursuing a clinical trial for the app with a pulmonologist at the Fox Chase Cancer Center.
 - Feature-extraction models based on the physical attributes of the nodules correlated to areas and location of high intensity pixels.
 - Screening for other pulmonary disorders: Model training can include other disorders, providing comprehensive diagnostics in a universal system.
 - Explore the idea of visual LLMs (vLLM) for higher fidelity & explainability.